

Real and Complex Restricted-Boltzmann-Machine Neural Quantum States for Heisenberg Models in a D-Wave-Assisted Variational Monte Carlo Scheme

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Abstract

The D-Wave-assisted method of Gardas, Rams, and Dziarmaga is a variational Monte Carlo method in which the quantum annealer is used as a sampler for a real Boltzmann distribution associated with a restricted Boltzmann machine (RBM), not as a direct simulator of the target quantum Hamiltonian. Consequently, a real-valued RBM in that construction represents a positive wave-function amplitude in the computational basis. Such an ansatz is appropriate for Heisenberg-type Hamiltonians only when the ground state can be made nonnegative by a fixed basis choice or by a known sign transformation. This includes ferromagnetic XXZ/Heisenberg models in the standard S^z basis, and unfrustrated bipartite antiferromagnets after the Marshall sign transformation or an equivalent staggered spin rotation. Frustrated antiferromagnets, nonbipartite antiferromagnets, Dzyaloshinskii-Moriya interactions, complex boundary twists, and generic sign-incompatible fields require a sign or phase ansatz. A D-Wave quantum processing unit cannot directly sample a complex-valued RBM because its programmable problem Hamiltonian is a real binary quadratic model. It can, at most, sample a real amplitude model while the sign or phase is evaluated classically in the local energy and in the stochastic-reconfiguration gradients.

I. PURPOSE AND NOTATION

This note addresses the following questions. Which Heisenberg models can be treated with the real-valued RBM ansatz used in the D-Wave-assisted approach of Gardas, Rams, and Dziarmaga [1]? Can a D-Wave quantum annealer sample complex RBMs? Are there constraints on the Heisenberg couplings or fields when the sampler is D-Wave, Gibbs sampling, or simulated annealing? Finally, what changes are needed to learn Heisenberg ground states rather than the transverse-field Ising ground states considered in Ref. [1]?

I use two distinct notations to avoid a common ambiguity. The target Heisenberg couplings and fields are denoted by K_{ij}^α and g_i^α , with $\alpha = x, y, z$. The D-Wave or classical binary-quadratic-model coefficients are denoted by J_{ij}^{BQM} and h_i^{BQM} . These are not the same objects. In the Gardas scheme, the target Hamiltonian enters through the variational local energy, whereas the sampler is programmed with the RBM energy.

For spin-1/2 variables, let $v_i = \pm 1$ denote the eigenvalue of $2S_i^z$ in the computational

basis. A general real XYZ model with one-body fields may be written as

$$H_{\text{XYZ}} = \sum_{\langle ij \rangle} (K_{ij}^x S_i^x S_j^x + K_{ij}^y S_i^y S_j^y + K_{ij}^z S_i^z S_j^z) + \sum_i (g_i^x S_i^x + g_i^y S_i^y + g_i^z S_i^z). \quad (1)$$

The isotropic Heisenberg model is the special case $K_{ij}^x = K_{ij}^y = K_{ij}^z \equiv K_{ij}$, usually with $g_i^x = g_i^y = 0$ when a longitudinal magnetic field is present. The XXZ model has $K_{ij}^x = K_{ij}^y \equiv K_{ij}^\perp$ and K_{ij}^z independent.

II. WHAT THE GARDAS CONSTRUCTION SAMPLES

The RBM ansatz used in Ref. [1] writes a visible-hidden energy

$$\phi_\theta(v, h) = a \cdot v + b \cdot h + h^T W v, \quad (2)$$

where $v_i, h_\mu \in \{\pm 1\}$ and $\theta = (a, b, W)$. With real parameters, Gardas *et al.* use

$$\psi_\theta(v) = \left[\sum_h \exp[-\phi_\theta(v, h)] \right]^{1/2}. \quad (3)$$

This is strictly nonnegative. Its probability distribution is

$$p_\theta(v) = \frac{|\psi_\theta(v)|^2}{\sum_{v'} |\psi_\theta(v')|^2} = \frac{\sum_h \exp[-\phi_\theta(v, h)]}{Z_\theta}. \quad (4)$$

Therefore a sampler for the joint positive Boltzmann distribution

$$p_\theta(v, h) = Z_\theta^{-1} \exp[-\phi_\theta(v, h)] \quad (5)$$

can be used as the importance sampler for variational Monte Carlo (VMC). This is the central observation of Ref. [1]: the annealer samples the RBM distribution, and the classical part of the algorithm computes the stochastic-reconfiguration covariance matrix, the forces, and the local energy.

The target Hamiltonian is not embedded on D-Wave. In Ref. [1] the target model is the transverse-field Ising Hamiltonian, and the D-Wave chip is used to sample the RBM probability. The paper emphasizes that the annealer measures only in the computational basis and therefore does not directly measure the target transverse-field-Ising energy. The same distinction is essential for Heisenberg models.

If the annealer were an exact equilibrium sampler, one would identify the RBM parameters with the annealer's Ising coefficients up to an effective inverse temperature β :

$$h_i^{\text{BQM}} \sim \frac{a_i}{\beta}, \quad h_\mu^{\text{BQM}} \sim \frac{b_\mu}{\beta}, \quad J_{\mu i}^{\text{BQM}} \sim \frac{W_{\mu i}}{\beta}. \quad (6)$$

In practice, Ref. [1] replaces β by an estimated value β_x because the effective sampling temperature of a D-Wave run is device-, schedule-, and coefficient-dependent.

III. THE SIGN STRUCTURE OF HEISENBERG GROUND STATES

A. The stoquastic criterion

A real positive ansatz such as Eq. (3) can represent the exact ground state only if that ground state can be chosen nonnegative in the basis in which v is sampled, or if a known sign factor has been separated out. A sufficient and widely used condition is stoquasticity: in a chosen basis all off-diagonal matrix elements of the Hamiltonian are real and nonpositive. For an irreducible stoquastic Hamiltonian, the Perron-Frobenius theorem implies that the ground-state vector may be chosen strictly positive in that basis [4–6].

This is why the real RBM construction works naturally for the transverse-field Ising Hamiltonian

$$H_{\text{TFIM}} = -\Gamma \sum_i \tau_i^x - \sum_{\langle ij \rangle} J_{ij}^z \tau_i^z \tau_j^z, \quad \Gamma \geq 0, \quad (7)$$

because the off-diagonal single-spin-flip matrix elements are $-\Gamma$ in the τ^z basis. The same criterion can be applied to Heisenberg models.

For the XXZ Hamiltonian

$$H_{\text{XXZ}} = \sum_{\langle ij \rangle} \left[K_{ij}^z S_i^z S_j^z + \frac{K_{ij}^\perp}{2} (S_i^+ S_j^- + S_i^- S_j^+) \right] + \sum_i g_i^z S_i^z, \quad (8)$$

the only off-diagonal two-spin matrix elements in the S^z basis are $K_{ij}^\perp/2$, connecting $|\uparrow_i \downarrow_j\rangle$ and $|\downarrow_i \uparrow_j\rangle$. Thus, in the unrotated S^z basis, the model is stoquastic whenever

$$K_{ij}^\perp \leq 0 \quad \text{for every active exchange bond } (ij). \quad (9)$$

This includes ferromagnetic XXZ and ferromagnetic isotropic Heisenberg couplings. The diagonal parameters K_{ij}^z and g_i^z do not affect stoquasticity, although of course they affect the ground state.

A local staggered rotation by π about the z axis changes the sign of S_i^x and S_i^y on selected sites. Equivalently, choose signs $\eta_i = \pm 1$ and transform

$$S_i^\pm \mapsto \eta_i S_i^\pm, \quad S_i^z \mapsto S_i^z. \quad (10)$$

The off-diagonal exchange coefficient on bond (ij) becomes $\eta_i \eta_j K_{ij}^\perp/2$. Therefore an XXZ model with longitudinal fields is sign-free in some local z -rotation gauge if and only if there exists an assignment $\eta_i = \pm 1$ such that

$$\eta_i \eta_j K_{ij}^\perp \leq 0 \quad \text{for all bonds } (ij). \quad (11)$$

For uniform antiferromagnetic $K_{ij}^\perp > 0$, this is possible exactly on a bipartite interaction graph. It is the basis-rotation version of the Marshall sign rule for unfrustrated bipartite antiferromagnets [7, 8]. In the original spin basis the wave function carries a known sign, often written as

$$\psi(v) = (-1)^{N_{\downarrow, B}(v)} A(v), \quad A(v) \geq 0, \quad (12)$$

where B is one sublattice. The positive amplitude $A(v)$ may be represented by the real RBM of Eq. (3), while the sign in Eq. (12) must be included in the local energy.

B. Classification of Heisenberg-type models for a real Gardas RBM

With the above criterion, the answer is as follows.

1. **Correct without an additional phase/sign network: ferromagnetic XXZ or Heisenberg models in the S^z basis.** These satisfy $K_{ij}^\perp \leq 0$ on all exchange bonds. Arbitrary K_{ij}^z and longitudinal fields g_i^z remain diagonal and do not spoil positivity. The isotropic ferromagnet has a degenerate ground-state multiplet, so in numerical work it is often useful to choose a fixed magnetization sector or add a small longitudinal field to select one representative ground state.
2. **Correct after a known sign transformation: bipartite unfrustrated anti-ferromagnets.** For $K_{ij}^\perp > 0$ on a bipartite graph and with no sign-incompatible transverse terms, a staggered rotation or the Marshall sign factor makes the amplitude nonnegative. The real RBM samples $A(v)^2$, not the signed wave function itself. The sign must be used when evaluating $\psi(v')/\psi(v)$ in the local energy.

3. **Correct for more general signed XXZ graphs only when Eq. (11) is satisfied.**

This includes mixed ferromagnetic and antiferromagnetic couplings if the signed graph is unfrustrated in the sense that a local η_i gauge exists. A simple all-antiferromagnetic triangle fails this test.

4. **Not correctly represented by the positive real Gardas ansatz in general:**

frustrated antiferromagnets and nonbipartite antiferromagnets. Triangular-lattice antiferromagnets, kagome antiferromagnets, J_1 - J_2 square-lattice antiferromagnets in the frustrated regime, and spin-liquid candidates generally require a nontrivial sign structure. The positive ansatz may still be used as a variational approximation, but it is not an unbiased representation of the exact ground-state manifold.

5. **Generally not representable by a positive real RBM: models with intrinsic complex phases.**

Dzyaloshinskii-Moriya interactions, y -fields, complex boundary twists, and chiral spin terms produce imaginary or phase-dependent off-diagonal matrix elements unless they can be removed by a special gauge transformation. These require a complex phase ansatz or a known analytic phase.

6. **XYZ models require additional sign checks.**

For a bond in an XYZ Hamiltonian, the exchange part can be written as

$$K_{ij}^x S_i^x S_j^x + K_{ij}^y S_i^y S_j^y = \frac{K_{ij}^x + K_{ij}^y}{4} (S_i^+ S_j^- + S_i^- S_j^+) + \frac{K_{ij}^x - K_{ij}^y}{4} (S_i^+ S_j^+ + S_i^- S_j^-). \quad (13)$$

A single $\eta_i \eta_j$ gauge changes the sign of both coefficients. Therefore a positive real ansatz is adequate only when both classes of off-diagonal matrix elements can be made nonpositive on every bond. If the two coefficients have incompatible signs on the same bond, a sign-free real amplitude-only treatment is not available in that basis.

The practical rule is therefore not “Heisenberg versus Ising”, but rather “sign-free amplitude versus nontrivial sign or phase”. A real RBM of the Gardas type is an amplitude model. It becomes a full wave-function model only when the sign or phase is trivial or known in advance.

IV. CAN D-WAVE SAMPLE A COMPLEX RBM?

D-Wave quantum processing units accept real Ising or QUBO coefficients and return samples of binary variables. Their user-programmable problem is a real binary quadratic model on the hardware working graph, possibly after minor embedding. They do not accept complex local fields, complex couplings, or complex Boltzmann weights.

Consider a complex RBM wave function

$$\psi_\theta(v) = \sum_h \exp[-\phi_R(v, h) - i\phi_I(v, h)]. \quad (14)$$

The sampling distribution needed for VMC is the positive distribution

$$p_\theta(v) = \frac{|\psi_\theta(v)|^2}{\sum_{v'} |\psi_\theta(v')|^2}. \quad (15)$$

Although Eq. (15) is positive, it is not generally the marginal of a real pairwise RBM energy. The modulus square contains interference between hidden configurations,

$$|\psi_\theta(v)|^2 = \sum_{h, h'} \exp[-\phi_R(v, h) - \phi_R(v, h')] \exp[-i(\phi_I(v, h) - \phi_I(v, h'))], \quad (16)$$

which is not a positive Boltzmann sum term by term. For the analytically marginalized form $\psi(v) = \exp(a \cdot v) \prod_\mu 2 \cosh(b_\mu + W_\mu \cdot v)$ with complex parameters, $|\psi(v)|^2$ contains factors $|\cosh(x + iy)|^2$ whose logarithm is not, in general, a binary quadratic energy in the visible and hidden variables. Thus a complex RBM cannot be sampled directly by programming complex RBM parameters on D-Wave.

There is one useful restricted alternative. Write the neural quantum state as an amplitude-phase product

$$\psi_{\theta, \varphi}(v) = A_\theta(v) \exp[i\chi_\varphi(v)], \quad A_\theta(v) \geq 0, \quad (17)$$

where the amplitude distribution $A_\theta(v)^2$ is represented by a real RBM marginal of the form Eq. (4). Then D-Wave can be used to sample the amplitude distribution, while the phase network $\chi_\varphi(v)$ is evaluated classically in the local energy and in the gradients. For a known real sign, this reduces to $\chi(v) = 0$ or π as in the Marshall sign rule.

Therefore the answer is precise: *D-Wave cannot sample a complex RBM as a complex Boltzmann machine. It can only sample a real positive auxiliary distribution.* For complex Heisenberg ground states, the D-Wave part must be restricted to the amplitude sampler, and the complex phase must be handled outside the annealer.

V. LOCAL ENERGY FOR HEISENBERG MODELS

To adapt the VMC part of Ref. [1], replace the transverse-field-Ising local energy by the Heisenberg local energy. Let $v^{(i)}$ be the configuration obtained from v by flipping spin i , and let $v^{(ij)}$ be the configuration obtained by flipping both spins i and j . For the Hamiltonian in Eq. (1), the local energy is

$$\begin{aligned}
E_{\text{loc}}(v) &= \frac{\langle v | H | \psi \rangle}{\psi(v)} \\
&= \sum_{\langle ij \rangle} \frac{K_{ij}^z}{4} v_i v_j + \sum_i \frac{g_i^z}{2} v_i \\
&\quad + \sum_{\langle ij \rangle} \left[\frac{K_{ij}^x + K_{ij}^y}{4} \mathbf{1}_{v_i = -v_j} + \frac{K_{ij}^x - K_{ij}^y}{4} \mathbf{1}_{v_i = v_j} \right] \frac{\psi(v^{(ij)})}{\psi(v)} \\
&\quad + \sum_i \left(\frac{g_i^x}{2} + \frac{i g_i^y v_i}{2} \right) \frac{\psi(v^{(i)})}{\psi(v)}. \tag{18}
\end{aligned}$$

The last term assumes the sign convention of Eq. (1); it should be changed if the magnetic-field term is written with an overall minus sign. For the usual XXZ model with only a longitudinal field, Eq. (18) reduces to

$$E_{\text{loc}}^{\text{XXZ}}(v) = \sum_{\langle ij \rangle} \frac{K_{ij}^z}{4} v_i v_j + \sum_i \frac{g_i^z}{2} v_i + \sum_{\langle ij \rangle: v_i = -v_j} \frac{K_{ij}^\perp}{2} \frac{\psi(v^{(ij)})}{\psi(v)}. \tag{19}$$

For the isotropic Heisenberg model, set $K_{ij}^\perp = K_{ij}^z = K_{ij}$.

If the wave function has a known sign or learned phase,

$$\psi(v) = A(v) \exp[i\chi(v)], \tag{20}$$

then every off-diagonal ratio must be evaluated as

$$\frac{\psi(v')}{\psi(v)} = \frac{A(v')}{A(v)} \exp\{i[\chi(v') - \chi(v)]\}. \tag{21}$$

This is the main mathematical change from the real transverse-field-Ising implementation. The probability distribution for sampling remains $A(v)^2/Z$, but the sign or phase appears in the estimator.

VI. PARAMETER CONSTRAINTS FOR D-WAVE AND FOR CLASSICAL SAMPLERS

A. Physical Heisenberg parameters

In a VMC implementation of the Gardas type, the physical Heisenberg couplings K_{ij}^α and fields g_i^α are not programmed into the D-Wave chip. They are used in Eq. (18). Therefore there is no D-Wave hardware range constraint on the physical K_{ij}^α or g_i^α themselves. One may rescale the entire target Hamiltonian by a positive constant without changing the ground-state vector, although this changes the numerical scale of the energy estimator and can affect conditioning and stochastic-reconfiguration step sizes.

There are, however, theoretical constraints on using a positive real ansatz. The off-diagonal exchange and field terms must be sign-free in the chosen basis, or a sign/phase factor must be supplied. Those are constraints on the sign structure of the target Hamiltonian, not on D-Wave hardware ranges.

B. D-Wave sampler parameters

The D-Wave constraints apply to the RBM sampler coefficients h_i^{BQM} and J_{ij}^{BQM} after the RBM has been mapped to an Ising or QUBO problem. The relevant restrictions are:

1. **Problem type.** The QPU accepts real Ising or QUBO binary quadratic models. It does not accept Heisenberg exchange operators or complex RBM coefficients as target Hamiltonian terms.
2. **Coefficient ranges.** The allowed h^{BQM} and J^{BQM} ranges are solver properties. For example, D-Wave documentation lists typical Advantage values such as $h_{\text{range}} = [-4, 4]$, $J_{\text{range}} = [-1, 1]$, and $\text{extended_j_range} = [-2, 1]$, while the announced Advantage2 system has a nominal bias range $[-6, 6]$. The exact ranges should be queried through the solver properties for the QPU used.
3. **Connectivity and embedding.** A dense RBM visible-hidden graph generally requires minor embedding. Chain strengths consume part of the available coupling scale, can introduce chain breaks, and can bias the sampled distribution. This issue was already central in Ref. [1], which embedded RBMs on Chimera hardware.

4. **Effective temperature.** Scaling all RBM coefficients changes the sampled distribution unless the effective inverse temperature is adjusted accordingly. In Eq. (6), the correct scale is set by an effective β that is not known a priori. The Gardas algorithm compensates by adapting an estimate β_x .
5. **Precision and noise.** Very small coefficients are vulnerable to analog control noise, while very large coefficients may force rescaling and reduce the resolution of smaller terms. This is an implementation accuracy issue rather than a Heisenberg-model parameter restriction.

Thus the physical Heisenberg parameters enter classically, whereas the neural-network sampling parameters must fit the D-Wave binary-quadratic hardware and embedding constraints.

C. Gibbs, Metropolis, and simulated annealing

Classical samplers remove most hardware constraints, but not the sign-structure issue of the wave-function ansatz.

For a real RBM amplitude, block Gibbs sampling is natural because the conditional distributions factorize:

$$p(h_\mu = 1 \mid v) = \frac{1}{1 + \exp[2(b_\mu + W_\mu \cdot v)]}, \quad (22)$$

$$p(v_i = 1 \mid h) = \frac{1}{1 + \exp[2(a_i + W_i^T h)]}, \quad (23)$$

up to the sign convention used in Eq. (2). Gibbs sampling has no hardware coefficient range and no minor-embedding overhead. Its limitation is mixing: large weights, multimodal distributions, and constrained magnetization sectors can lead to long autocorrelation times.

Metropolis-Hastings sampling on visible configurations works for real or complex neural quantum states because it only needs ratios of $|\psi(v')|^2/|\psi(v)|^2$. It is therefore the standard fallback for complex RBMs. The phase is not sampled as a probability; it appears in Eq. (21).

Simulated annealing samples a real binary quadratic model or seeks its low-energy states. Like Gibbs sampling, it has no analog hardware range, although numerical floating-point precision and the chosen inverse-temperature schedule matter. It is an approximate sam-

pler unless it is equilibrated carefully at the desired distribution. It can be useful for the amplitude RBM but does not solve the complex-phase problem by itself.

VII. REQUIRED CHANGES TO THE GARDAS APPROACH FOR HEISENBERG GROUND STATES

A Heisenberg implementation should modify the original transverse-field-Ising workflow as follows.

A. Choose a basis and phase convention

First determine whether the Heisenberg Hamiltonian is sign-free after a local gauge transformation. For XXZ models with longitudinal fields, solve Eq. (11). If it has a solution, train the positive amplitude in that rotated basis. For a bipartite antiferromagnet, this is equivalent to using the Marshall sign factor in Eq. (12).

If no such transformation exists, do not use Eq. (3) as the full wave function. Use either

$$\psi(v) = s_{\text{known}}(v)A_{\theta}(v), \quad s_{\text{known}}(v) = \pm 1, \quad (24)$$

when the sign is known, or a learned amplitude-phase ansatz

$$\psi(v) = A_{\theta_A}(v) \exp[i\chi_{\theta_x}(v)]. \quad (25)$$

In the latter case, the D-Wave sampler can only sample $A_{\theta_A}(v)^2$ if this amplitude is encoded as a real RBM. The phase network is evaluated on a classical processor.

B. Replace the local-energy estimator

The transverse-field-Ising local energy used by Gardas *et al.* must be replaced by Eq. (18) or by the XXZ simplification Eq. (19). The off-diagonal exchange terms flip two spins, not one spin, for the $S_i^+ S_j^- + S_i^- S_j^+$ part. If pair-flip XYZ terms or transverse fields are included, the relevant one-spin and two-spin ratios must be added with the correct coefficients and phases.

C. Use complex stochastic reconfiguration when phases are learned

For a real positive amplitude, the stochastic-reconfiguration equations are essentially the same as in Ref. [1]. For a complex ansatz, the logarithmic derivatives

$$O_k(v) = \frac{1}{\psi(v)} \frac{\partial \psi(v)}{\partial \theta_k} \quad (26)$$

are complex. The covariance matrix is Hermitian,

$$S_{kl} = \langle O_k^* O_l \rangle - \langle O_k^* \rangle \langle O_l \rangle, \quad (27)$$

and the force vector is

$$F_k = \langle E_{\text{loc}} O_k^* \rangle - \langle E_{\text{loc}} \rangle \langle O_k^* \rangle. \quad (28)$$

The update can be written in terms of complex parameters, or in terms of real and imaginary parts using Wirtinger derivatives. The energy minimized remains real, up to sampling error.

D. Respect conserved sectors when useful

The isotropic and XXZ Heisenberg models with only longitudinal fields conserve total S^z . For antiferromagnetic even-size systems, the ground state is usually sought in the $S_{\text{tot}}^z = 0$ sector. Sampling the full 2^N configuration space is correct but inefficient if the ansatz or the Hamiltonian should remain in a fixed sector. One should either use magnetization-conserving proposals in a classical sampler, add a strong amplitude penalty for configurations outside the desired sector, or design a sector-constrained neural state. D-Wave can impose a quadratic magnetization penalty, but this increases embedding overhead; it is often simpler to enforce the sector in a classical visible-space sampler.

E. Validate sampler bias

D-Wave samples are approximate and depend on embedding, chain strength, anneal schedule, and effective temperature. For Heisenberg VMC this matters because inaccurate samples bias both the covariance matrix and the force vector. A practical implementation should compare D-Wave samples with classical Gibbs or Metropolis samples on small systems, monitor autocorrelation and chain-break statistics, and where possible use reweighting or calibration to estimate systematic error.

VIII. SUMMARY ANSWER

The Gardas approach can be extended to Heisenberg models, but not by directly replacing the transverse-field-Ising Hamiltonian with a Heisenberg Hamiltonian on the annealer. The annealer samples a real RBM probability distribution. A real-valued RBM of the Gardas form is a positive amplitude model. It is correct for Heisenberg ground states only when the target ground state is nonnegative in the chosen basis, or when a known sign transformation such as the Marshall sign has been factored out. This includes ferromagnetic XXZ/Heisenberg models and unfrustrated bipartite antiferromagnets after a staggered rotation. Frustrated, nonbipartite, chiral, or intrinsically complex Heisenberg models require a sign or phase ansatz.

D-Wave cannot directly sample a complex RBM. It can sample only a real Ising/QUBO distribution. For complex neural quantum states, the viable hybrid strategy is to use D-Wave, if at all, for a real amplitude distribution and to compute the sign or phase contribution classically in the local energy and gradients.

There are no D-Wave hardware constraints on the physical Heisenberg K and g parameters in a VMC calculation, because those parameters enter the local-energy estimator. The hardware constraints apply to the RBM sampling coefficients and their embedding. Classical Gibbs, Metropolis, and simulated-annealing samplers avoid analog hardware ranges, but they still require an appropriate positive sampling distribution and adequate mixing. The essential modification of the Gardas algorithm is therefore an ansatz modification, not merely a sampler substitution: one must add a sign or phase sector when the Heisenberg model is not sign-free, and replace the Ising local energy by the Heisenberg local-energy estimator.

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